

Trend Following: Replication, Practitioner Rules and Tail Protection

Part I: Lempérière, Deremble, Seager, Potters & Bouchaud (2014), “Two centuries of trend following”, reproduced on free data and tested after publication.

Part II: the core model of A. Clenow’s *Following the Trend* (2013) against the paper’s signal, on daily futures.

Part III: Dao et al. (2016), “Tail protection for long investors: trend convexity at work”, and trend as an overlay on an inverse-volatility portfolio

Liangxi Wang · monthly data 1960-01 → 2026-09 · daily futures 1990-01 → 2024-03 · October 2026
github.com/lwang-genomics/trend-following-replication

Contents

1	Summary	2
2	The paper and what is replicated	3
3	Data	3
4	Three data checks	3
5	Replication, 1960–2013	5
6	The signal saturates	7
7	After publication: 2014–2026	8
8	150 years of US data	9
9	Part II: practitioner rules on daily futures	9
10	Part III: trend convexity and tail protection	16
11	Conclusions and limitations	21
12	Appendix: code and reproducibility	22

1 Summary

Lempérière et al. (2014, Capital Fund Management) show that a minimal trend-following rule earns significant excess returns across commodities, currencies, stock indices and bonds: a t-statistic of about 5.9 on futures since 1960, and about 10 on spot data back to 1800. This note re-runs their rule on the same 27-market universe using only free public data, checks how the data affect the result, and then asks the question a published result invites: **did it hold after publication?**

- **The result replicates.** For 1960–2013 the aggregate Sharpe ratio is 0.74 (paper: 0.78), with $t = 5.4$ (5.7) and a drift-removed $t^* = 4.5$ (5.0). The trend P&L is only 0.18 correlated with the long-only drift (paper: 0.15). Currencies and bonds match the paper closely. Commodities are much weaker on spot prices (Sharpe 0.17 vs 0.80), as the paper itself anticipates, because spot data miss the futures' carry.
- **Saturation replicates.** A tanh fit of the next move on the signal saturates at $s^* = 1.01$ (paper: 0.89), beats the linear fit ($F = 11.2$), and the cubic term is negative, as reported.
- **After publication the effect is weaker.** Over 2014–2026 the Sharpe ratio is 0.27 ($t = 1.0$), against 0.74 in-sample. A block bootstrap of the in-sample months puts the probability of an out-of-sample Sharpe this low at 9%. So the result is weaker but not statistically rejected, and every n gives the same picture.
- **Two data traps are documented and avoided.** Most free series are **monthly averages**. Trading them the very next month manufactures a fake trend (Working, 1960). The paper's equation, which holds the position one month later, removes the artefact, and on markets with month-end data it reproduces the month-end result exactly (Sharpe 0.22 vs 0.22). **Administered prices** (grain support, oil price controls) need a stale-price filter; without one, a single corn observation from 1972 (-467σ) cuts the 1960–2013 Sharpe ratio from 0.82 to 0.34.
- **150 years of US data** (Shiller, from 1873) give a trend Sharpe of 0.45 with $t = 5.5$, positive in every 50-year block.

Part II takes the practitioner's side. Clenow's *Following the Trend* argues that a simple, diversified trend model with a moving-average filter, breakout entries, a trailing stop and volatility sizing captures much of what professional trend followers earn, and that the exact rule matters less than diversification and risk sizing. The model is run on 62 daily back-adjusted futures (1990 to March 2024, net of costs) next to the paper's signal on the same data.

- **Both work, and they are mostly the same bet.** Net Sharpe ratios for 1990–2013 are 0.75 (Clenow) and 0.86 (paper's EMA, 100 days $\approx$ 5 months), and 0.57 and 0.48 for 2014–2024. Their monthly returns are 0.83 correlated, and neither has a significant alpha over the other ($t = 0.4$ and 1.8).
- **The stop changes the shape, not the edge.** Clenow's model is in the market 48% of the time. At equal volatility its worst drawdown is 21% against 28%, and its monthly skew is 0.84 against 0.36.
- **The parameters do not matter much, and cannot be tuned.** Across 30 breakout and stop settings, the net Sharpe ratio ranges from 0.51 to 0.91 before 2014 and from 0.37 to 0.78 after. The in-sample ranking does not predict the later one (rank correlation -0.20).
- **A real trend fund.** The two rules together explain 62% of the monthly variance of a public managed-futures fund (AQR Managed Futures Strategy (AQMIX), 2010–2024).
- **Part I's weak commodities are mostly a data effect.** On the same seven commodities and months, the paper's monthly rule earns a Sharpe ratio of 0.19 on averaged spot prices and 0.72 on month-end futures. Most of the gap comes from the one-month delay that averaged data force, not from carry.

Part III tests the follow-up paper by the same group, Dao et al. (2016). It argues that trend following is a cheap form of tail protection for long investors because its P&L is the difference between long-term and short-term realised variance. It then puts that protection to work on the inverse-volatility portfolio of the companion study inverse-vol-futures-overlay.

- **The mechanics replicate exactly.** The paper's identity holds to machine precision on S&P 500 futures, and the P&L of a linear trend aggregated over ≈ 90 days is a parabola in the trend indicator ($R^2 = 0.96$, curvature 0.068 vs 0.067 predicted). The sign rule gives the predicted V .
- **Convexity in a real fund is weaker than in the paper.** A diversified replicator reaches a monthly correlation of 0.79 with a public managed-futures fund at the paper's $\tau = 180$ days. Measured at the right

horizon, the fund's convexity against the S&P 500 rises from $R^2 = 0.04$ to 0.10, less than the paper's $0.02 \rightarrow 0.18$ on the SG CTA Index.

- **The risk-parity bound holds on every day** (5434 days, 16 futures), as eq. (24) guarantees.
- **As an overlay, trend improves the inverse-vol book at equal risk.** At 10% volatility, adding the diversified trend raises the Sharpe ratio from 0.66 to 0.99, cuts the worst drawdown from 28% to 21% and turns quarterly skew from -0.71 to 0.12. In 2022, when the book lost 27%, the trend overlay made 32%. It does not, on average, pay off in the book's worst quarters.
- **Puts protect sooner but cost more.** The CBOE 5% put-protection index lost 3.7% a year against the S&P 500. Implied variance exceeded the variance realised afterwards in 85% of months. Puts helped most in the fast COVID crash, where a 180-day trend helped little.

Part I: Section 2 to Section 8. Part II: Section 9 to Section 9.7. Part III: Section 10 to Section 10.6. Conclusions: Section 11.

2 The paper and what is replicated

The paper defines, for each market and an exponential moving average (EMA) time scale of n months,

$$s_n(t) = \frac{p(t-1) - \langle p \rangle_{n,t-1}}{\sigma_n(t-1)}, \quad Q_n(t) = \sum_{t' < t} \text{sign}[s_n(t')] \frac{p(t'+1) - p(t')}{\sigma_n(t'-1)},$$

where $\langle p \rangle_{n,t}$ is an EMA of past prices excluding $p(t)$ and σ_n is an EMA of absolute monthly price changes. The position is long or short one unit of risk, with no costs. The paper uses monthly closes, $n = 5, 27$ markets (indices, 10-year bonds and currencies of seven countries, plus seven commodities) and Global Financial Data. Its aggregate P&L is the sum of the risk-normalised P&Ls; Sharpe ratios need no risk-free rate because futures are self-financed; $t = \text{Sharpe} \times \sqrt{\text{years}}$; and the “de-biased” t^* is the t -stat of the intercept after regressing the trend P&L on the long-only P&L.

Two implementation details. (i) As printed, eq. (2) pairs the signal built from $p(t'-1)$ with the price change $p(t') \rightarrow p(t'+1)$, so the position is held one month after the information date; this replication follows the equation literally ($\text{lag} = 1$). (ii) An EMA “with decay rate n months” is implemented with weight $\frac{1}{n}$ on the newest observation, and a 24-month warm-up precedes the first position.

3 Data

Sector	Free source (paper: Global Financial Data)	From
Indices (7)	OECD total share price indices (FRED SPASTT01..M661N)	1955–60
Bonds (7)	OECD 10-year yields (FRED IRLTLT01..M156N), turned into a 10-year bond price by exact monthly repricing	1953–69; Japan 1989
Currencies (6)	Fed exchange rates vs USD (FRED); Deutsche mark spliced to the euro in 1999	1973 (end of Bretton Woods)
Commodities (7)	World Bank “Pink Sheet”: maize, US HRW wheat, sugar, beef (live-cattle proxy), copper, US natural gas; WTI crude (FRED); IMF series extend the Pink Sheet	1960; natural gas 1985; crude 1982
US long history	Shiller: S&P composite and 10-year rate	1871; bond 1953

Start dates follow the paper's rule that only freely traded, moving prices can trend. Currencies start after Bretton Woods; crude starts after US oil price controls ended in 1981 (the price was flat in 63 of 96 months in 1974–81); natural gas starts after wellhead decontrol.

4 Three data checks

4.1 Monthly averages fake a trend unless the position is lagged

The OECD, Fed and World Bank monthly series are averages of daily prices. Averaging a random walk gives its monthly changes a lag-1 autocorrelation of 0.25 (Working, 1960), and every series here shows 0.25–0.35. A rule that trades the month right after the signal harvests that artefact. For the markets where free daily data exist (six currencies and the US 10-year), Table 1 compares averaged and month-end prices directly.

Market	From	AC(1) avg	AC(1) month-end	SR avg, lag 0	SR avg, lag 1	SR month-end, lag 0	SR month-end, lag 1
FX_GBP	1973	0.33	0.06	0.45	0.12	0.22	0.23
FX_JPY	1973	0.32	0.05	0.69	0.37	0.38	0.31
FX_CAD	1973	0.26	-0.06	0.52	0.28	0.08	0.10
FX_CHF	1973	0.26	0.01	0.49	0.06	0.21	0.02
FX_AUD	1973	0.31	0.01	0.45	0.15	0.12	0.09
FX_EUR	1999	0.30	0.04	0.65	0.30	0.30	0.12
BD_US	1962	0.30	0.09	0.58	0.28	0.22	0.10
Mean				0.55	0.22	0.22	0.14

Table 1: Monthly averages vs month-end closes, $n = 5$. AC(1): lag-1 autocorrelation of monthly log changes. Lag 0 trades the next month; lag 1 is eq. (2) as printed.

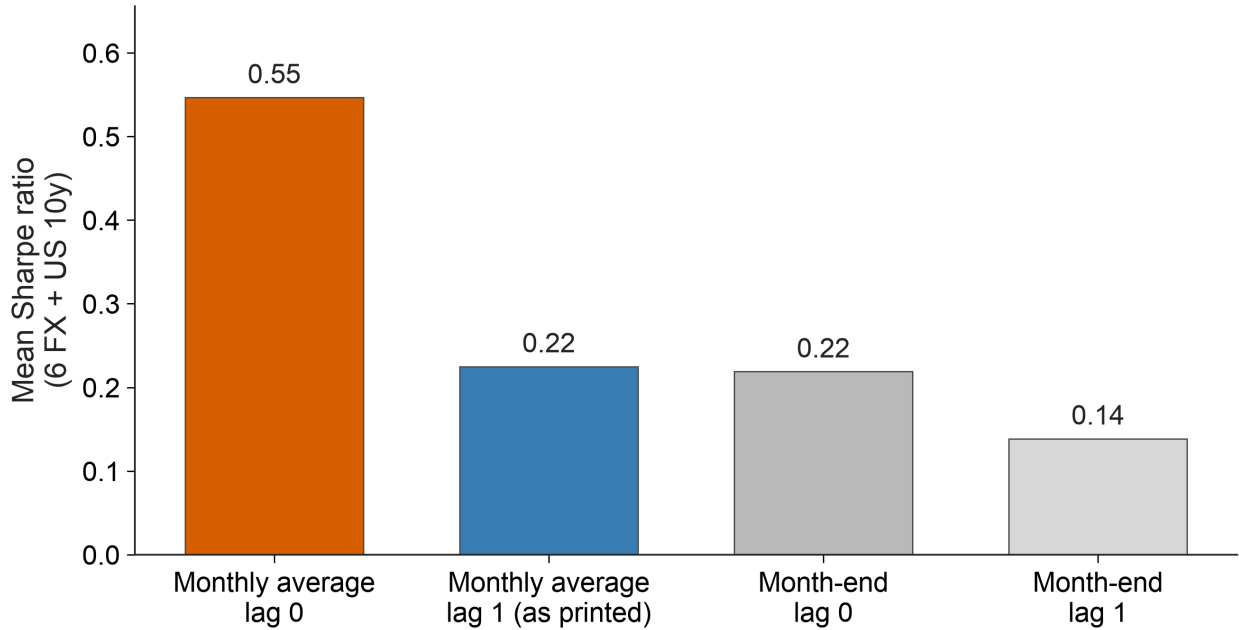


Figure 1: Mean Sharpe ratio over the seven markets in Table 1.

Month-end prices have no autocorrelation (≈ 0.03). On averaged data, lag 0 more than doubles the Sharpe ratio (0.55), a pure artefact; a simulation in the test suite shows that the same rule earns a mean t-stat of about 2 on averaged **random walks**. Lag 1 removes the artefact, and its Sharpe on averaged data (0.22) matches the realistic month-end, lag-0 strategy (0.22). The main specification (averaged data, lag 1) is therefore both faithful to the paper and realistic. The cost is that very short time scales, whose information decays within a month, look weaker than in the paper (Table 2).

4.2 Administered prices need a stale-price filter

Grain prices in the 1960s–70s were set by support programmes and often unchanged for months. σ_n then collapses towards zero, and the first real move is divided by almost nothing: corn in January 1972 contributes -467σ . Without a filter, this single observation pulls the 1960–2013 Sharpe ratio from 0.82 to 0.34. The paper requires “prices actually moving (no gaps)”. Here a market is traded in a month only if its price changed in at least 10 of the previous 12 months. The rule uses only past data, and the result hardly depends on the threshold (Sharpe 0.74, 0.74 and 0.70 for 10, 11 and 12 of 12).

4.3 Interpolated history is not data

Shiller’s long-term rate before 1953 is interpolated from annual observations. Its monthly changes have a lag-1 autocorrelation of 0.92 (0.31 afterwards), which handed the US bond a spurious trend Sharpe of 2.0 in 1873–1899. The long-history bond therefore starts in 1953.

5 Replication, 1960–2013

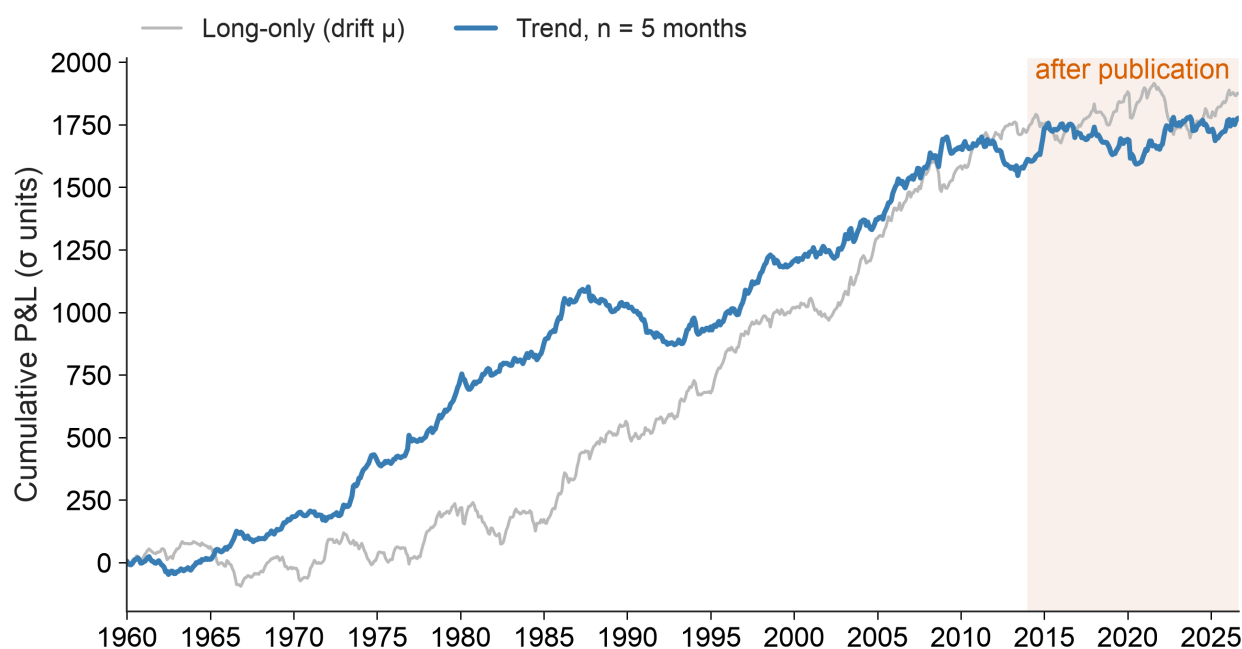


Figure 2: Aggregate trend P&L ($n = 5$, σ units, summed over markets) and the long-only drift. Shaded: after publication.

n (months)	SR paper	SR ours	t paper	t ours	t* paper	t* ours	SR 2014–26	t 2014–26
2	0.80	0.39	5.9	2.9	5.5	2.8	0.21	0.7
3	0.83	0.65	6.1	4.8	5.5	4.0	0.13	0.5
5	0.78	0.74	5.7	5.4	5.0	4.5	0.27	1.0
7	0.80	0.70	5.9	5.1	5.0	4.1	0.26	0.9
10	0.76	0.64	5.6	4.7	5.1	3.4	0.30	1.1
15	0.65	0.58	4.8	4.3	4.5	2.4	0.21	0.8
20	0.57	0.53	4.2	3.9	3.3	1.9	0.23	0.8

Table 2: Paper's Table 1 vs this replication, 1960–2013, plus the out-of-sample Sharpe ratio. t*: drift-removed t-stat.

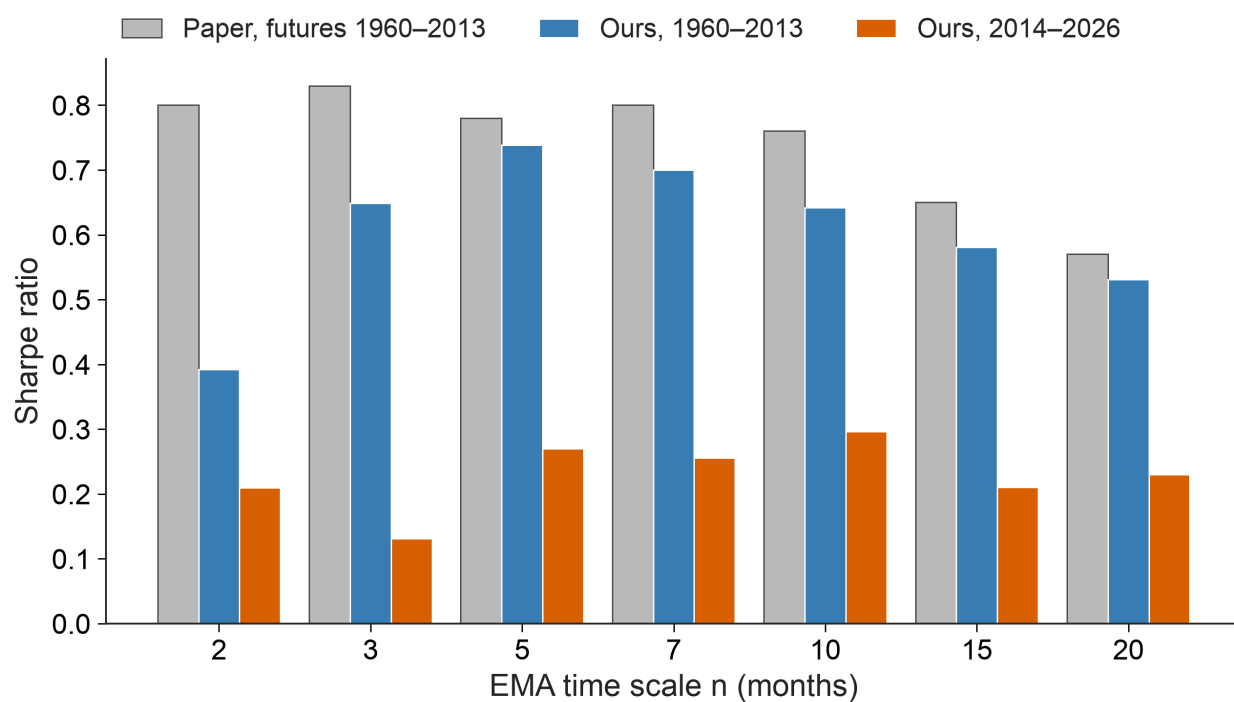


Figure 3: Sharpe ratio by EMA time scale: paper (futures), replication (spot proxies), and after publication.

Sector	SR paper	SR ours	t paper	t ours	t* paper	t* ours	SR(μ) ours	Start paper	Start ours	SR 2014–26	t 2014–26
Commodities	0.80	0.17	5.9	1.2	5.0	-0.4	0.69	01/1960	03/1962	0.32	1.2
Currencies	0.57	0.53	3.6	3.3	3.4	3.2	0.16	05/1973	05/1975	-0.02	-0.1
Bonds	0.49	0.45	2.8	3.3	1.6	3.3	0.04	05/1982	01/1960	0.20	0.7
Indices	0.41	0.57	2.3	4.2	2.1	2.6	0.68	01/1982	01/1960	0.13	0.5

Table 3: Paper's Table 2 vs this replication by sector, $n = 5$, through 2013, with the 2014–2026 Sharpe ratio. The paper's bond and index futures start in 1982; the spot proxies start around 1960.

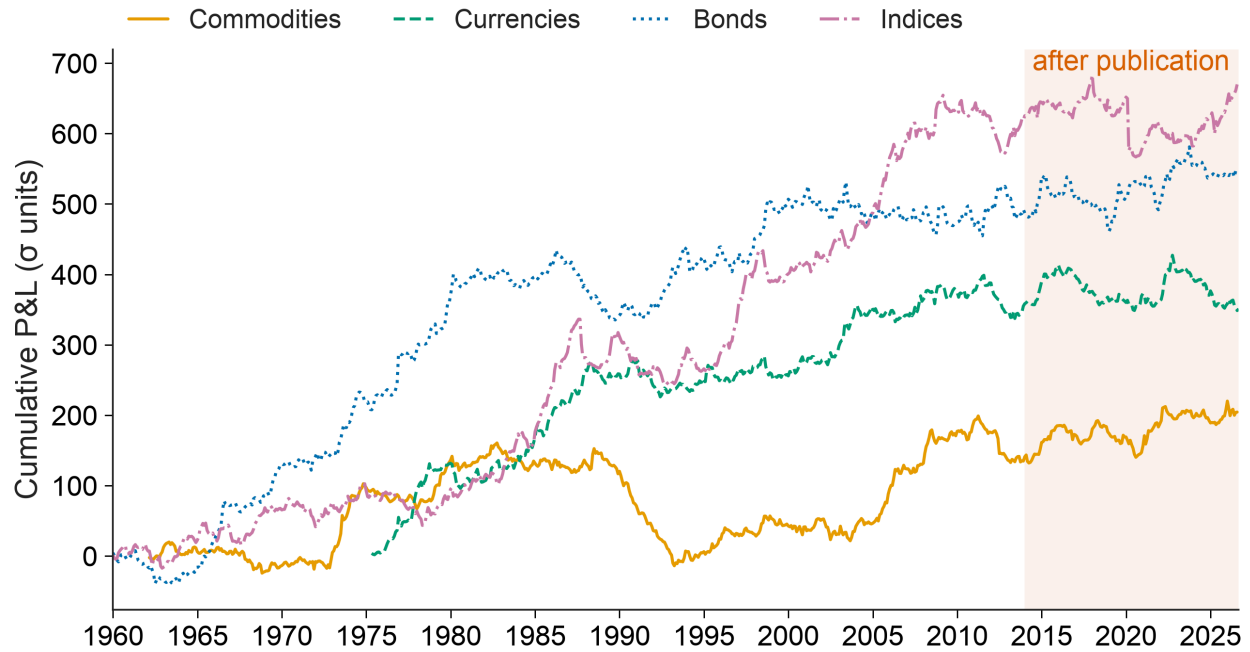


Figure 4: Cumulative trend P&L by sector.

Period	SR paper	SR ours	t paper	t ours	t* paper	t* ours	SR(μ) ours	t(μ) ours
1960s	0.66	0.77	2.1	2.5	1.8	2.4	-0.08	-0.3
1970s	1.15	1.46	3.6	4.6	2.5	4.8	0.53	1.7
1980s	1.05	0.79	3.3	2.5	2.9	2.1	0.64	2.0
1990s	1.12	0.42	3.5	1.3	3.0	0.3	1.00	3.2
2000–13	0.75	0.57	2.8	2.1	1.9	1.0	1.00	3.8
2014–26	–	0.27	–	1.0	–	0.8	0.22	0.8

Table 4: Paper's Table 3 vs this replication by decade, $n = 5$; the last row is after publication.

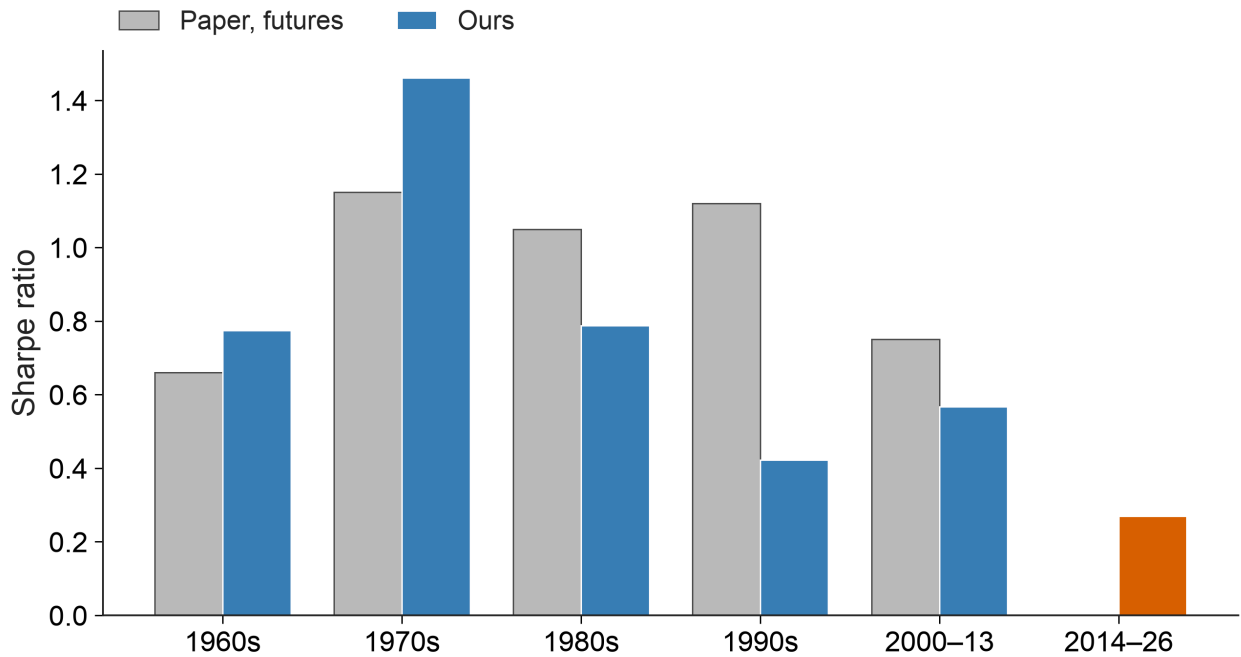


Figure 5: Sharpe ratio by decade. Orange: after publication.

The aggregate replication is close for $n \geq 5$: Sharpe 0.53–0.74 vs 0.57–0.80. Every decade from the 1960s to 2013 is positive. Sector by sector, currencies (0.53 vs 0.57) and bonds (0.45 vs 0.49) match; indices are stronger on the longer spot history. Commodities are the clear gap (0.17 vs 0.80). The paper reports only a 65% spot–futures correlation for commodities, because futures add a carry (roll-yield) term that spot prices lack. Averaged monthly commodity prices, and beef as a stand-in for live cattle, add noise. Part II (Section 9.7) tests this on futures: most of the gap comes from the one-month delay that averaged data force, not from carry.

6 The signal saturates

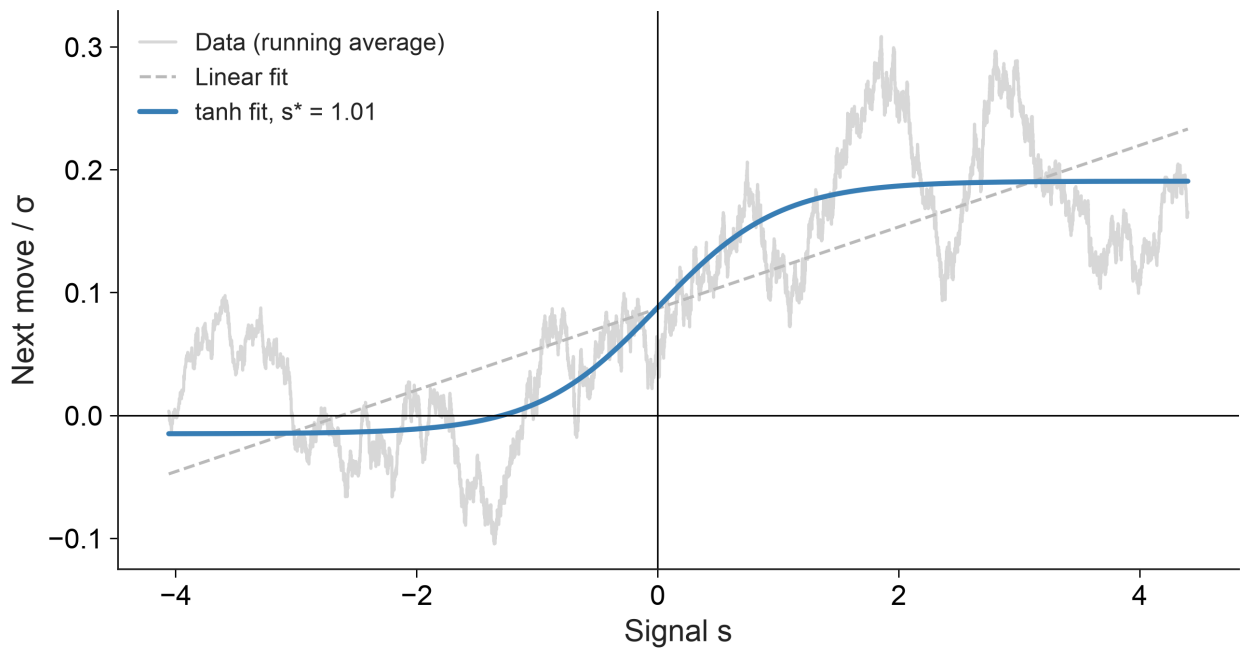


Figure 6: Next normalised move vs signal, pooled over markets ($n = 5$, 18992 points): running average over 1,000 points, linear fit and tanh fit.

Fit	a	b	s*	Note
Linear, paper	0.018	0.038	–	
Linear, ours	0.087	0.033	–	
tanh, paper	–	0.075	0.89	
tanh, ours	0.088	0.101	1.01 ± 0.61	F vs linear = 11.2

The predictive slope flattens for strong signals. The tanh model is preferred to the linear one, the cubic coefficient is negative (-0.0023), and the saturation scale s^* is close to the paper's. The intercept is larger here because the spot proxies' drift is larger in σ units; the signal is noisy, so s^* is only loosely determined.

7 After publication: 2014–2026

Period	Years	Sharpe	t-stat	Long-only Sharpe
1960–2013 (in sample)	54	0.74	5.4	0.69
2014–2026 (after publication)	12.8	0.27	1.0	0.22

The aggregate Sharpe ratio fell from 0.74 to 0.27 after publication, and 5 of 13 calendar years were negative. Two tests put this in proportion:

- **Difference of Sharpe ratios:** $z = -1.51$, using the standard error $\sqrt{(1/Y_1 + 1/Y_2)}$.
- **Bootstrap:** drawing 12.75-year paths from the in-sample months (12-month blocks), an out-of-sample Sharpe at or below 0.27 occurs 9% of the time; the 90% range is 0.15 to 1.33.

So the post-publication period is unusually weak but not inconsistent with the historical effect. With a Sharpe ratio near 0.7, drawdowns of several years are expected: the paper itself notes that their typical length is about $1/S^2$ years. The weakness is broad, not one sector: indices 0.13, bonds 0.20, currencies -0.02 , commodities 0.32. 2022, the year of the inflation shock, was the best year since 2014. The long-only drift also weakened ($0.69 \rightarrow 0.22$), so the period was poor for simple directional strategies in general.

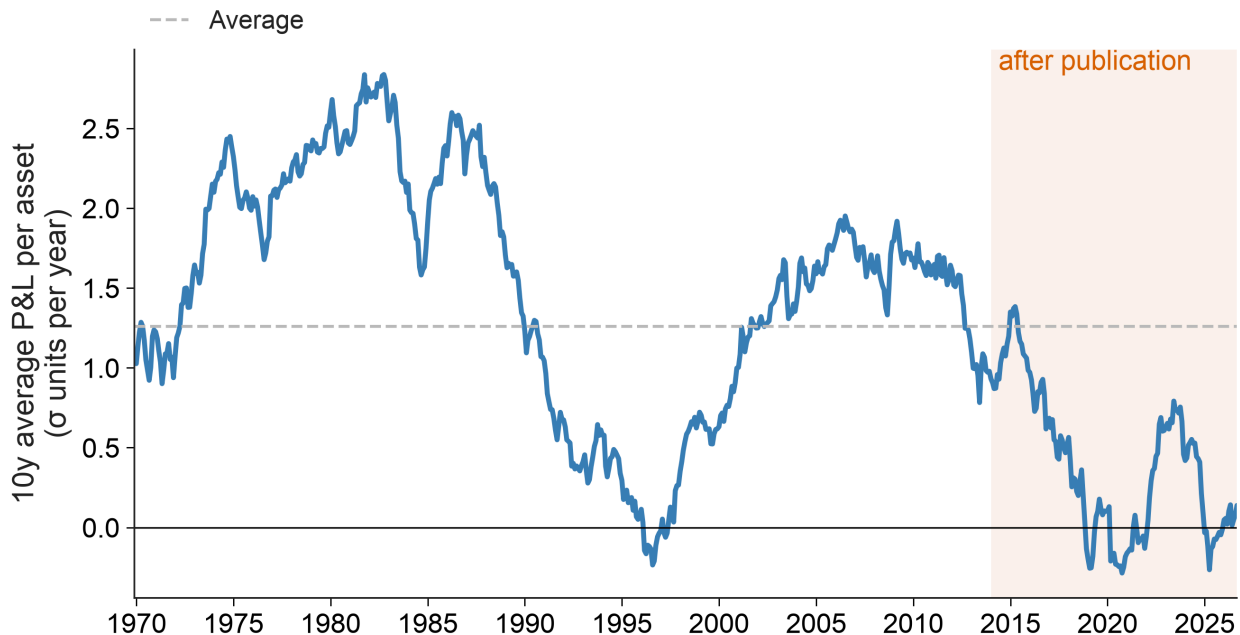


Figure 7: Trailing 10-year average trend P&L per market (σ units per year).

The paper reports that the 10-year performance of the trend was never negative in two centuries. On these free spot proxies it does dip below zero: briefly in 1996 (-0.29 at the low, 2020–10) and again around 2018–2021. So that claim does not replicate on this data, even before publication. The likely cause is the weaker commodity sector, which carried much of the paper's performance.

8 150 years of US data

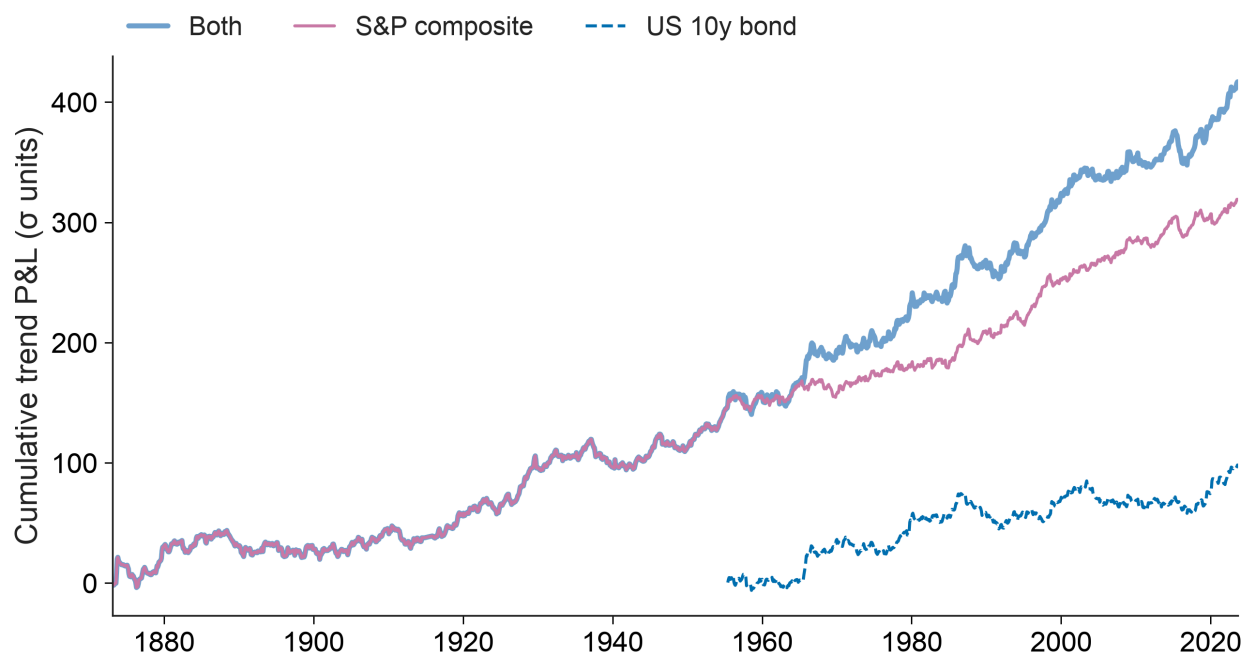


Figure 8: Cumulative trend P&L on the S&P composite (from 1873) and the US 10-year bond (from 1953), Shiller data.

Period	SR (2 assets)		t	t*	SR equity	SR bond
1873–2023	0.45	5.5	5.2		0.42	0.27
1873–1899	0.18	0.9	0.9		0.18	–
1900–1949	0.34	2.4	1.9		0.34	–
1950–1999	0.60	4.3	3.9		0.56	0.30
2000–2023	0.57	2.8	2.3		0.56	0.23

Table 5: Trend on US equity and bonds, Shiller data, $n = 5$, lag 1.

On US data alone, the trend has a Sharpe of 0.45 over 149 years ($t = 5.5$, drift-removed 5.2) and is positive in every 50-year block, consistent with the paper’s long-history finding.

9 Part II: practitioner rules on daily futures

Part I tested the academic version of trend following: one signal, monthly data, no costs. Practitioners trade something that looks different. The core model of Clenow’s *Following the Trend* (2013) trades daily, enters on breakouts, exits on trailing stops and sizes positions by volatility. The book’s thesis is that this kind of simple model, run on a diversified futures universe, captures much of what large trend-following funds earn. Part II asks three questions:

1. Is the practitioner’s rule better than, or different from, the paper’s signal on the same data?
2. What does each component (filter, breakout, stop) contribute, and do the parameters matter?
3. How closely do both track a real managed-futures fund?

9.1 Data, rules and costs

Futures. Daily back-adjusted (“Panama”) prices of 62 futures from the open-source pssystemtrade project, pinned to one commit (4420802541, data to 28 March 2024). Back-adjustment shifts history at each roll, so price **changes** are those of the contract held, roll yield included. All P&L is therefore computed in price points and converted to capital through risk-based sizing; no price level is used. Results start in January 1990 (34 markets), and later contracts join as their data begin.

Sector	Markets	First data	Last start	Cost (% ATR)	Markets (pysystemtrade names)
Equities	13	1982	2014	1.2	SP500, NASDAQ, DOW, FTSE100, DAX, CAC, AEX, IBEX, EUROSTX, SMI, HANG, NIKKEI, KOSPI
Rates	14	1978	2012	2.4	US2, US5, US10, US20, US30, BUND, BOBL, SHATZ, GILT, JGB, CAD10, BTP, OAT, EURIBOR
Currencies	8	1972	2003	1.9	EUR, JPY, GBP, CHF, AUD, CAD, NZD, MXP
Energy	6	1980	2006	1.5	CRUDE_W, CRUDE_ICE, HEATOIL, GASOILINE, GASOIL, GAS_US
Metals	5	1970	1995	1.8	GOLD, SILVER, COPPER, PLAT, PALLAD
Agriculturals	16	1969	2007	3.2	CORN, WHEAT, REDWHEAT, SOYBEAN, SOYMEAL, SOYOIL, SUGAR11, COFFEE, COCOA, COTTON2, OJ, RICE, OATIES, LIVECOW, FEEDCOW, LEANHOG

Table 6: Futures universe. Cost: median cost per contract and side, as a percentage of the daily ATR.

Clenow's core model (parameters as described in the book; `config.ClenowRules`):

- trend filter: long trades only while the 50-day EMA is above the 100-day EMA, short trades only while below;
- entry: a close at the highest (lowest) close of the last 50 days, in the filter's direction;
- exit: a trailing stop 3 ATRs from the best close since entry;
- size: 0.2% of capital per ATR(100), set at entry and not rebalanced.

Closes only. The data have no highs and lows, so the average true range is estimated from closes. On 28 front-month futures with daily OHLC (Yahoo Finance, 2000–2024), the 100-day true range averages 1.90 times the mean absolute close-to-close change (range 1.36 to 2.04, against 2 for a Brownian motion). ATR is therefore $1.9 \times \text{the mean } |\Delta p|$ over 100 days. Without this correction the “3 ATR” stop would be about half as wide as the book's.

The paper's signal on daily data: $\text{position} = \text{sign}(p - \text{EMA}_n[p]) / \sigma_n$, with σ_n the EMA of $|\Delta p|$, rebalanced daily, $n = 100$ days ($\approx$ the paper's 5 months), and the same risk per market as Clenow's model.

Components. Three partial models isolate the parts of the core model: *Filter only* (always positioned with the EMA filter), *Breakout only* (stop-and-reverse on 50-day highs and lows) and *Filter + breakout* (breakout entries, exit when the filter flips).

Execution and costs. Decisions use closes up to day t and are traded at the close of day $t+1$ (the book trades the next open). Costs per contract and side are half the bid-ask spread plus commission, from the same project's configuration. They are converted to a fraction of each market's recent ATR (median 2.1%) and charged at that fraction of the ATR at the time of each trade, so they scale with volatility through history. Every roll trades the position twice. These are today's costs, so results are also shown with costs tripled. EURIBOR's spread estimate in the source (0.26 points, about 50 ticks) is an evident error and is replaced by one tick. P&L is not compounded, and positions are fractional (a large account).

9.2 Practitioner rule vs the paper's signal

Rule	SR gross	SR net	SR net, costs $\times 3$	SR net 2014–24	Max DD (10% vol)	Skew (monthly)
Filter only	0.90	0.83	0.68	0.40	–39%	0.52
Breakout only	0.55	0.46	0.27	0.30	–37%	0.51
Filter + breakout	0.89	0.82	0.69	0.45	–35%	0.54
Clenow core	0.86	0.75	0.54	0.57	–21%	0.84
Paper EMA, 100 d	1.04	0.86	0.51	0.48	–28%	0.36

Table 7: Net Sharpe ratios by rule, 1990–2013 unless stated. Max drawdown and skew over 1990–2024, with each rule scaled to 10% annual volatility.

Rule	Vol (risk 0.2%)	Time in market	Changes per market-year	Open positions	Corr. with paper EMA	Corr. with Clenow
Filter only	41%	99%	8.6	49	0.87	0.72
Breakout only	37%	99%	9.8	49	0.81	0.84
Filter + breakout	41%	93%	8.9	46	0.86	0.72
Clenow core	23%	48%	10.4	24	0.83	1.00
Paper EMA, 100 d	32%	99%	15.5	49	1.00	0.83

Table 8: How the rules trade, 1990–2024. Vol: annual volatility at 0.2% risk per position. Changes: entries, exits and reversals. Open positions: average number. Correlations of monthly net returns.



Figure 9: Cumulative net P&L, each rule scaled to 10% annual volatility over 1990–2024 (display only). Shaded: after the paper and the book were published.

The paper's signal and Clenow's core model earn similar risk-adjusted returns: 0.86 and 0.75 net before 2014, and 0.48 and 0.57 after. A paired block bootstrap (63-day blocks) of the Sharpe difference, Clenow minus paper, gives -0.11 (95% interval -0.33 to 0.12) before 2014 and 0.09 (-0.30 to 0.43) after. Neither is distinguishable from zero.

Regression (monthly, net)	Period	t(alpha)	Appraisal ratio	R ²
Clenow core on paper EMA (100 d)	1990–2013	0.4	0.09	0.69
Clenow core on paper EMA (100 d)	1990–2024	1.0	0.17	0.68
Clenow core on 4 EMA horizons	1990–2013	1.3	0.27	0.77
Clenow core on 4 EMA horizons	1990–2024	1.7	0.29	0.76
Paper EMA (100 d) on Clenow core	1990–2013	1.8	0.39	0.69
Paper EMA (100 d) on Clenow core	1990–2024	1.5	0.25	0.68

Table 9: Spanning regressions of monthly net returns. Appraisal ratio: annualised alpha over residual volatility.

The two are largely the same bet. The paper's signal explains 69% of the monthly variance of Clenow's model, and four EMA horizons explain 77%. What remains has an appraisal ratio of 0.27 ($t = 1.3$), which is not significant. The reverse regression gives the same answer ($t = 1.8$).

EMA n (days)	SR gross	SR net	SR net 2014–24	Corr. with Clenow core
20 (1 month)	0.73	0.33	0.20	0.76
50 (2 months)	1.00	0.76	0.34	0.85
100 (5 months)	1.04	0.86	0.48	0.83
200 (10 months)	0.95	0.81	0.65	0.72

Table 10: The paper's signal on daily futures by EMA time scale.

On daily futures the paper's signal is strongest between 50 and 200 days, as in the paper's monthly Table 1. The fastest version (20 days) loses most of its edge to costs.

9.3 What each component does

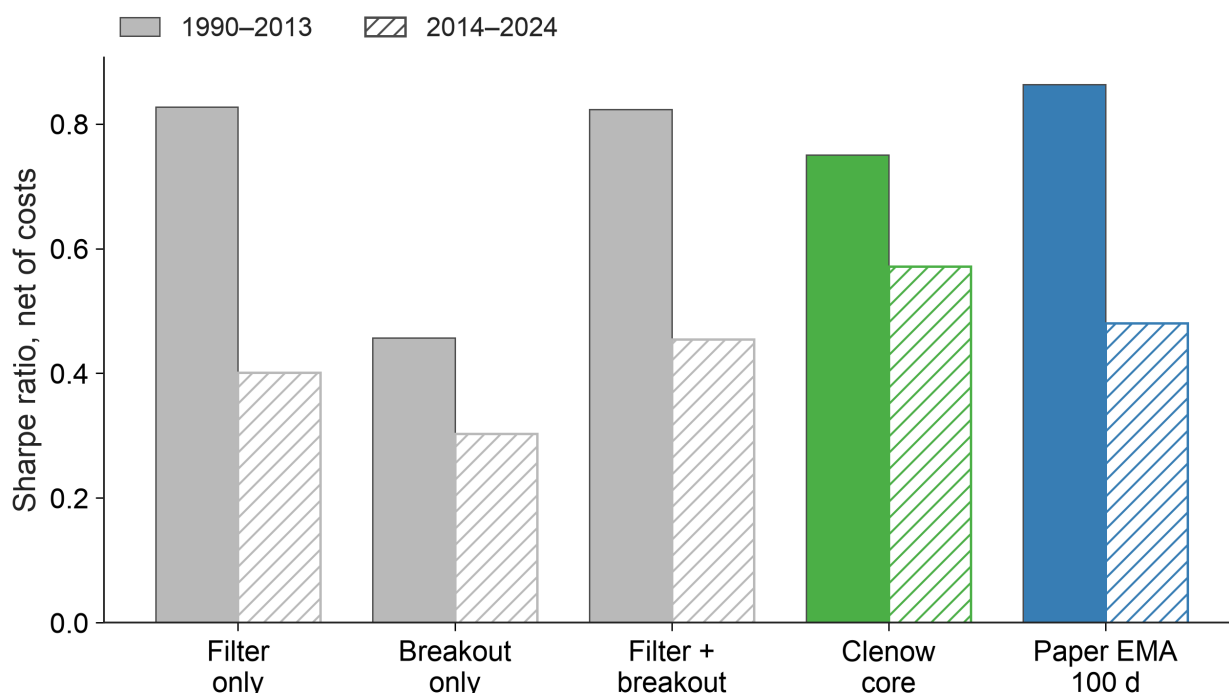


Figure 10: Net Sharpe ratio of each rule before and after 2014.

- **The filter carries the edge.** The 50/100-day EMA filter alone earns 0.83 before 2014, as much as anything else here. Breakouts alone earn less (0.46); adding them to the filter changes little (0.82).
- **The trailing stop changes the shape.** It keeps the model out of the market 52% of the time and cuts losing trades early. At equal volatility, the worst drawdown falls from 35% to 21%, and monthly skew rises from 0.54 to 0.84. The in-sample Sharpe ratio is a little lower, and with costs tripled it drops more than the filter's (0.54 vs 0.68).
- **After 2014** every rule is weaker. The stop-based model held up best (0.57), but the differences between rules are well within noise over ten years.

9.4 Do the parameters matter?

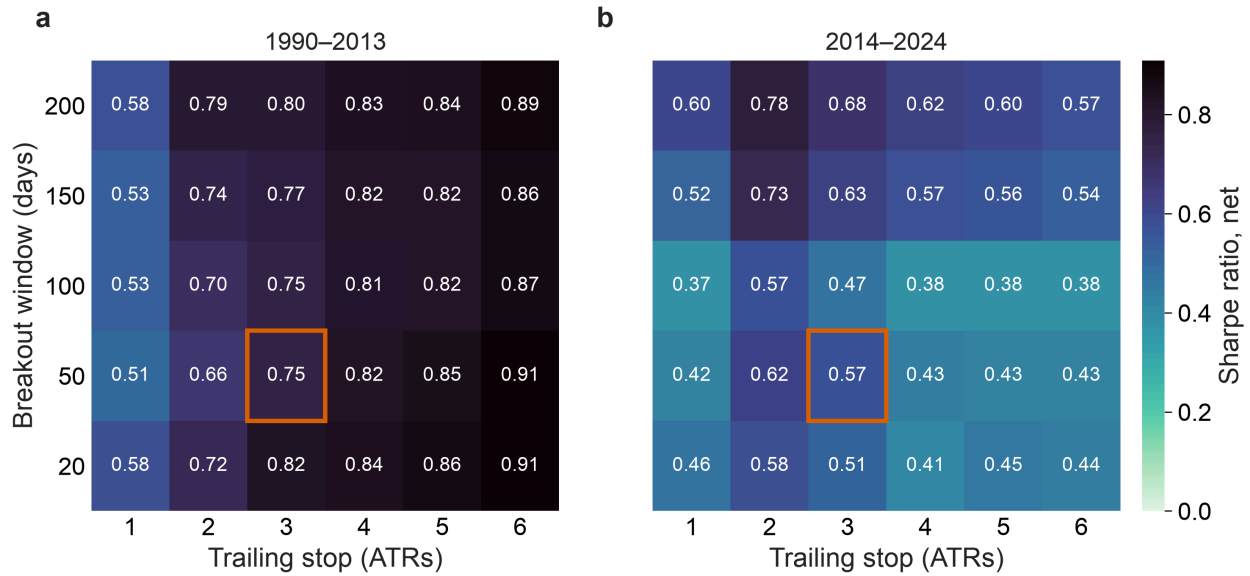


Figure 11: Net Sharpe ratio of the core model for breakout windows and trailing stops (in true-range ATRs); the filter stays at 50/100 days. Boxed: the book's setting.

All 30 settings are profitable in both periods: 0.51 to 0.91 before 2014 and 0.37 to 0.78 after. This supports the book's view that a reasonable trend model does not depend on fine-tuning. It also shows why tuning does not help. Before 2014 the widest stops look best; after 2014 the 2-ATR stop does, and the rank correlation between the two periods' grids is -0.20 . Choosing the in-sample optimum would not have improved the out-of-sample result.

9.5 Sectors and years

Sector	Markets	Clenow 1990–2013	Clenow 2014–24	Paper EMA 1990–2013	Paper EMA 2014–24
Equities	13	0.29	−0.04	0.44	0.19
Rates	14	0.44	0.57	0.61	0.63
Currencies	8	0.43	0.01	0.37	−0.11
Energy	6	0.36	0.83	0.44	0.42
Metals	5	0.33	−0.26	0.41	−0.27
Agriculturals	16	0.43	0.33	0.37	0.14

Table 11: Net Sharpe ratio by sector.

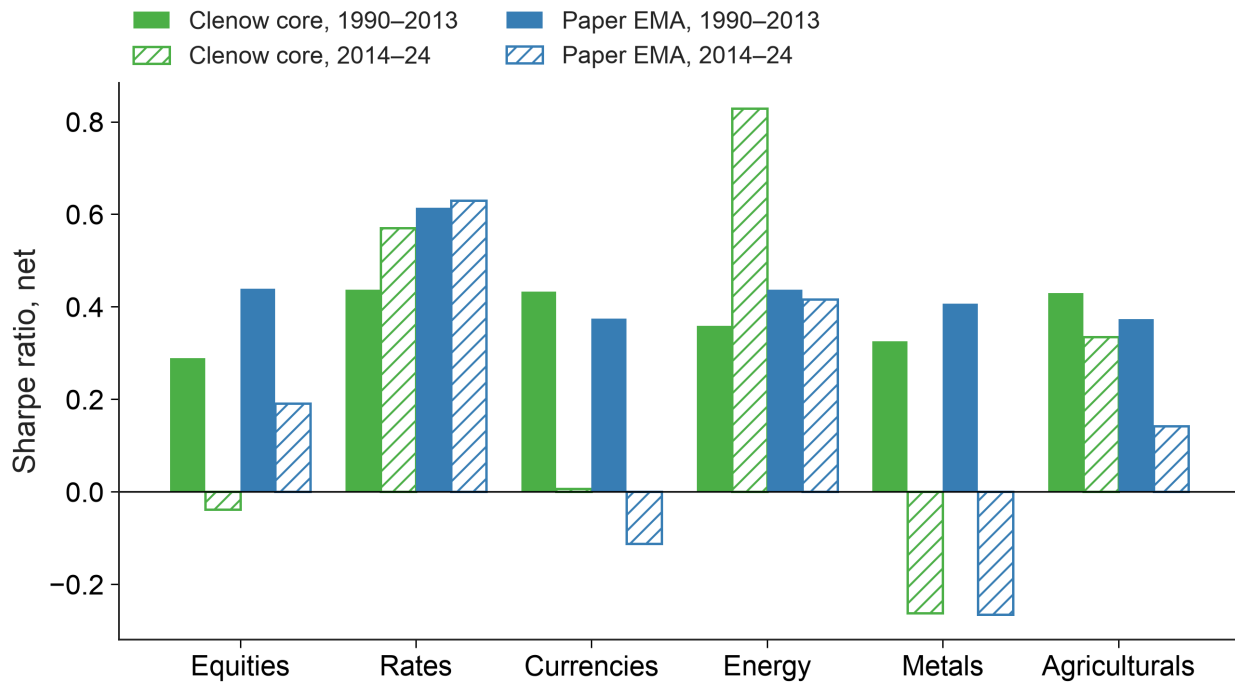


Figure 12: Net Sharpe ratio by sector, before and after 2014.

Every sector contributes before 2014. After 2014, rates and energy carry both rules, while metals and currencies lose. The sector pattern of the two rules is similar.

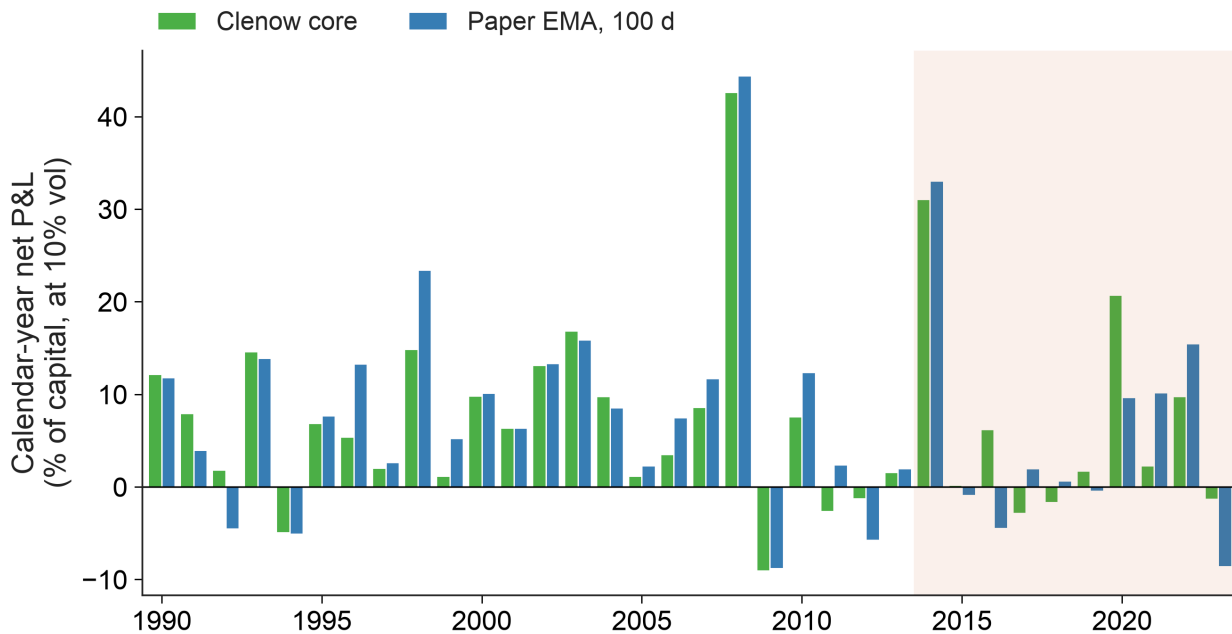


Figure 13: Calendar-year net P&L at 10% volatility, 1990–2023.

Year by year the two rules move together (correlation 0.91). They have 7 and 8 losing years out of 34, and both had their best year in 2008, when equity markets fell sharply.

9.6 A real trend fund

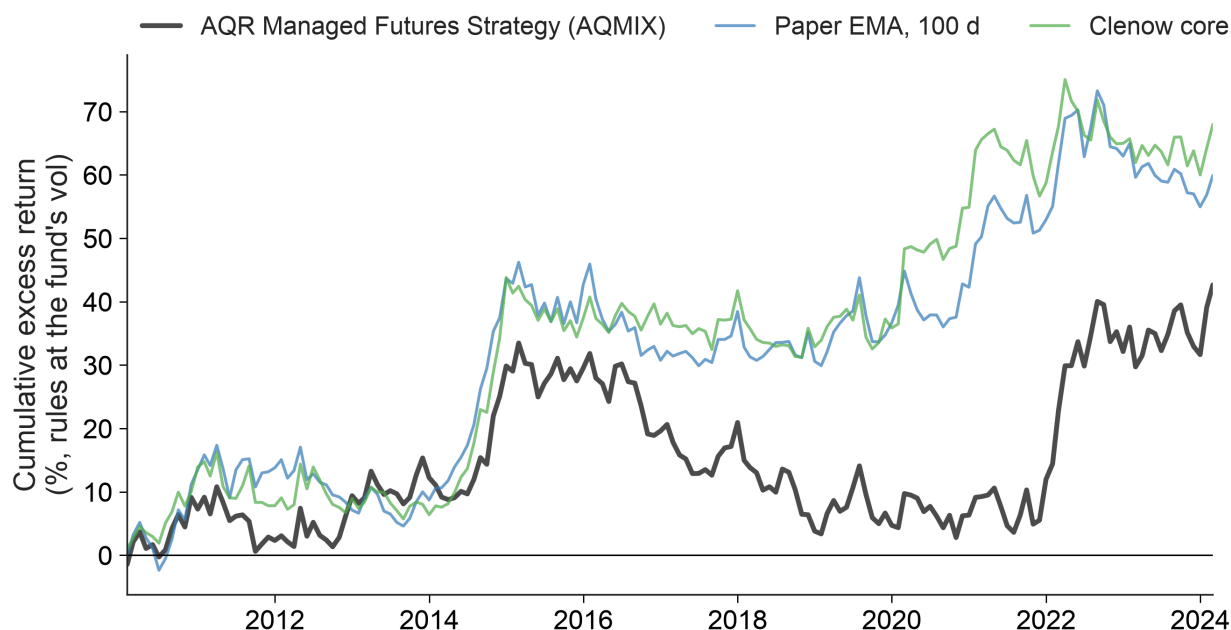


Figure 14: Monthly excess return over T-bills of a public managed-futures mutual fund (AQR Managed Futures Strategy (AQMIX), after fees), and the two rules scaled to the fund's volatility (before fees).

Clenow's thesis is that simple rules capture much of what professional trend followers do. As one test, the rules are compared with a public managed-futures mutual fund (AQR Managed Futures Strategy (AQMIX)), whose monthly total return (distributions reinvested, minus T-bills) is available from 2010-02 to 2024-03. Its returns are 0.70 correlated with Clenow's model and 0.78 with the paper's signal; together the two explain 62% of its monthly variance. The fund's Sharpe ratio over the period is 0.30 after fees, against 0.47 and 0.42 for the rules before fees and management costs. One fund is not the industry, but this one is largely explained by simple trend rules.

9.7 Back to Part I: are spot commodities the problem?

Part I found commodities far weaker on spot data than in the paper (Sharpe 0.17 vs 0.80) and attributed it to the futures' carry. Month-end futures prices allow a direct test: the paper's monthly rule ($n = 5$) on the seven Part I commodities, spot and futures side by side, over the same months of 1990–2013.

Commodity	Futures	From	Trend SR, spot (avg., lag 1)	Trend SR, futures (lag 1)	Trend SR, futures (lag 0)	Long-only SR, spot	Long-only SR, futures
Crude	CRUDE_W	1992	0.21	0.36	0.34	0.35	0.53
Natgas	GAS_US	1992	0.13	0.05	0.33	0.45	0.06
Corn	CORN	1990	0.11	-0.08	0.33	0.28	0.06
Wheat	REDWHEAT	1997	-0.12	0.16	0.42	0.46	0.03
Sugar	SUGAR11	1990	0.10	0.03	0.32	0.22	0.39
Cattle	LIVECOW	1990	-0.13	0.23	0.33	0.36	0.22
Copper	COPPER	1997	0.39	0.39	0.21	0.51	0.26
All seven			0.19	0.32	0.72	0.72	0.41

Table 12: Paper's monthly rule on spot (monthly averages, lag 1 as in Part I) and futures (month-end), same months up to 2013. Long-only: the drift term, lag 1.

On futures at the same lag, the seven-commodity trend Sharpe rises from 0.19 to 0.32. Month-end prices can be traded without the one-month delay that averaged data need (Part I, Table 1); doing so raises it to 0.72, close to the paper's 0.80 (measured on a longer period). The long-only drift is **lower** on futures (0.41 vs 0.72), which is the carry cost of holding commodities in contango. So the commodity shortfall of Part I is mostly due to the delay that monthly averages force on the rule, and only partly to carry. For currencies and bonds the delay costs nothing (Part I), so commodity trends seem to be faster. With 18–24 years per market these Sharpe ratios have standard errors of about 0.2, so the split is approximate.

10 Part III: trend convexity and tail protection

Dao, Nguyen, Deremble, Lempérière, Bouchaud and Potters (2016) explain why trend following tends to do well when markets move a lot. For a linear trend on one asset, they show that the P&L aggregated over the trend's horizon is, exactly, a long-term variance minus a short-term variance. A trend follower is therefore “long” large moves over its own time scale, like an option holder, and pays for it with short-term realised variance rather than an option premium. Part III checks the mathematics, measures the convexity in a real trend fund, and asks what the property is worth to a long-only investor. The investor in question holds the inverse-volatility portfolio of the companion study.

10.1 Trend P&L is long-term minus short-term variance

Following the paper, each market's daily price change D_t is normalised by a 10-day risk estimate, $R_t = D_t/\sigma_{t-1}$ with $\sigma_t = \gamma\sqrt{L_{10}[D_t^2]}$, where L_τ is an EMA with weight $2/(\tau+1)$. A linear trend holds $\Pi_t = \lambda\tau L_\tau[R_t]/\sigma_t$, so its daily P&L is $G_t = \lambda\tau L_\tau[R_t]_{t-1}R_t$. The paper's eq. (13) states

$$L_{\tau'}[G_t] = \frac{\lambda\tau}{\tau-1}(\tau L_\tau[R_t]^2 - L_{\tau'}[R_t^2]), \quad \tau' = \frac{\tau}{2} + \frac{1}{2\tau}.$$

With the trend indicator $T = \sqrt{\tau}L_\tau[R]$ and the P&L aggregated over $\approx \tau'$ days, $\bar{G} = \tau' L_{\tau'}[G]$, this reads $\bar{G} = \Upsilon(T^2 - L_{\tau'}[R^2])$: a parabola in the long-term move, minus the short-term variance. On S&P 500 futures the two sides agree to about 10^{-17} , the precision of the computer, and a test checks the identity on arbitrary fat-tailed series.

One detail differs from the paper. Its calibration factor $\gamma = 1.05$ makes R unit-variance; here $\gamma = 1.16$ is needed on S&P 500 futures (and 1.10 even on Gaussian returns). With the paper's value, $\langle R^2 \rangle = 1.22$, which shifts the parabola down but leaves its curvature unchanged. The theory curves below use the measured $\langle R^2 \rangle$.

10.2 The smile on S&P 500 futures

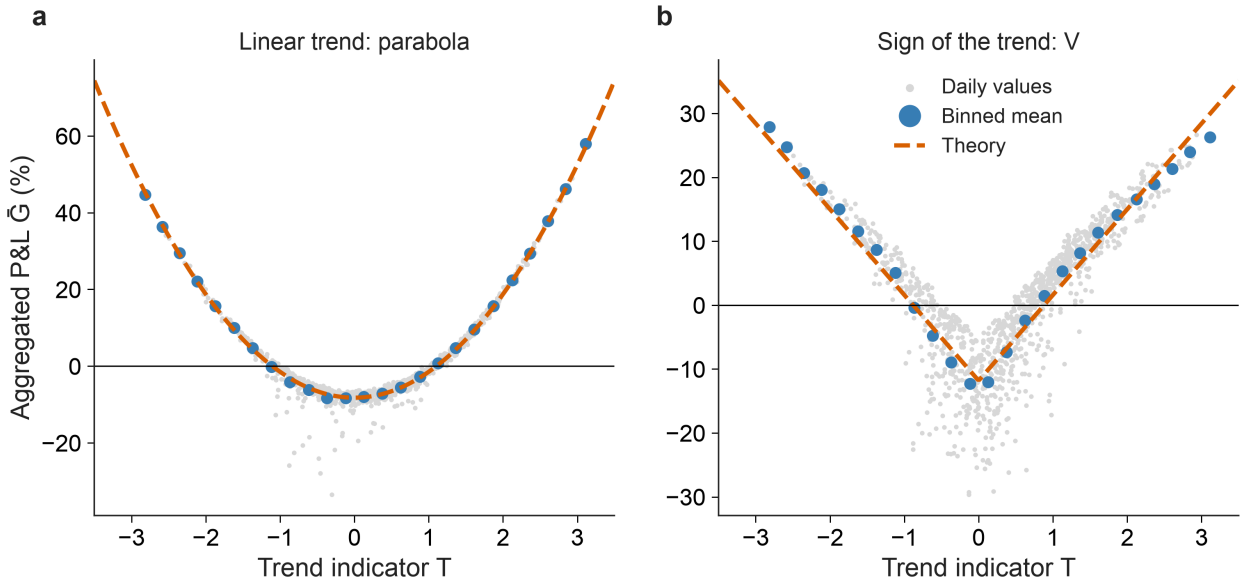


Figure 15: Aggregated P&L $\bar{G}$ (over $\tau' \approx 90$ days) of a trend on S&P 500 futures against the trend indicator T , $\tau = 180$ days, 1984-06 to 2015-10 (the paper's Figs. 4–5). (a) Linear trend, scaled to 1% daily P&L. (b) Sign of the trend. Dashed: theory.

The linear trend traces the predicted parabola: fitted curvature 0.0680 against 0.0675 from theory, intercept -0.084 against -0.082 , $R^2 = 0.96$. Capping the position at ± 1 turns the parabola into the predicted V (slope 0.148 vs 0.134, $R^2 = 0.80$). The continuous-time theory for the sign rule is approximate; the test suite checks it on simulated random walks.

Horizon (days)	Periods	Curvature c ($\times 100$)	R^2
1	10122	-0.05	0.23
5	2024	0.00	0.24
21	482	1.28	0.36
63	160	2.55	0.51
126	80	7.33	0.79
252	40	12.00	0.75

Table 13: Trend P&L on the S&P 500 against the S&P 500 move over the same non-overlapping periods, by horizon (linear trend, $\tau = 180$ days, 1984–2024). Quadratic fit: curvature c and R^2 .

The convexity only appears at the trend's own horizon. Measured day by day or week by week the curvature is nil; it grows with the horizon and dominates at three to twelve months (Table 13). A monthly scatter of trend P&L against S&P 500 returns gives $R^2 = 0.03$, which is why naive plots hardly show the effect.

10.3 Convexity in a real trend fund

The paper replicates the SG CTA Index with a linear trend on about 20 liquid futures, equal risk per market, and finds a broad maximum of correlation (above 80%) at $\tau \approx 180$ days. The SG CTA Index is not freely available, so the comparison here uses the public managed-futures fund of Part II (AQR Managed Futures Strategy (AQMIX), monthly excess returns since 2010). Two replicators are tested: the paper's list as available here (16 futures from 2002; Eurodollar and Short Sterling are missing, and the DAX stands in for the EuroStoxx 50) and the 62 futures of Part II.

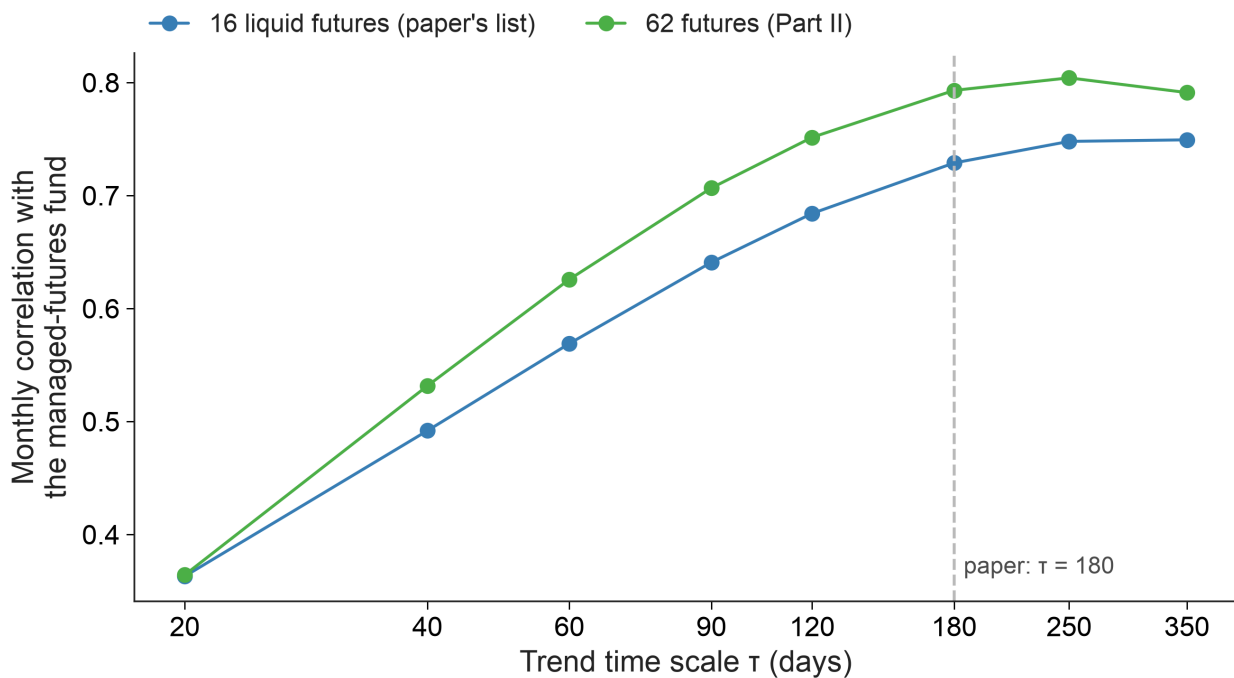


Figure 16: Monthly correlation between the replicator and the managed-futures fund (2010–2024) by trend time scale τ (the paper's Fig. 7).

The correlation rises with τ and flattens from about 180 days, at 0.79 for the 62-market replicator and 0.73 for the 16-market one. This is the same broad maximum as in the paper, at somewhat lower levels for a single fund.

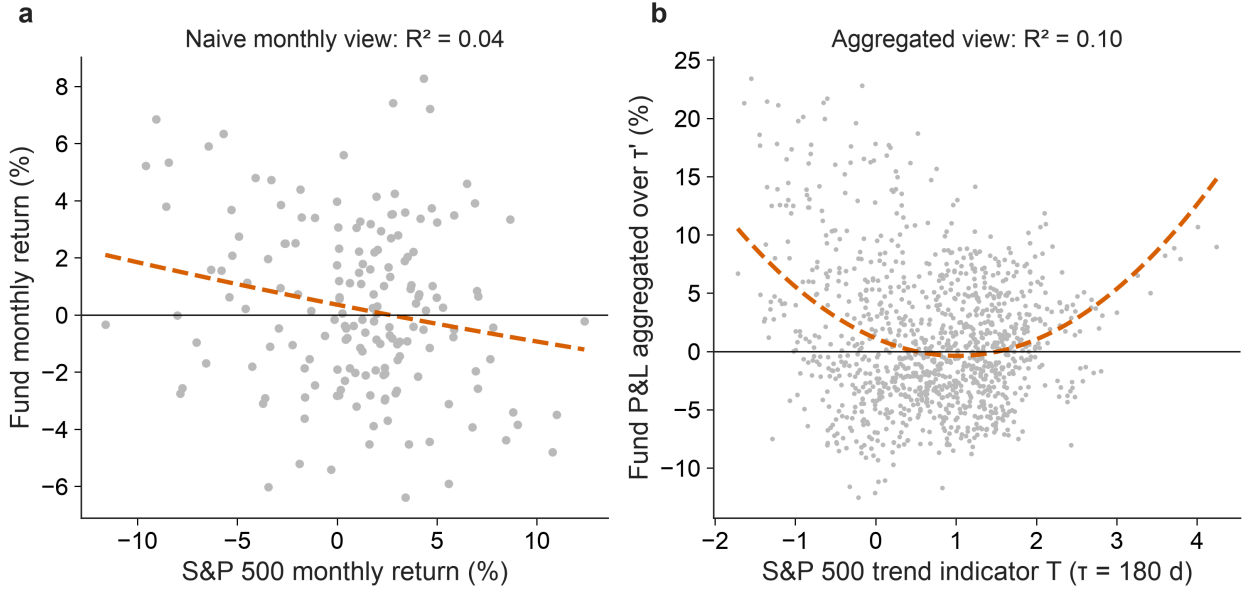


Figure 17: The fund against the S&P 500. (a) Monthly returns: the naive view (the paper's Fig. 2). (b) The fund's P&L aggregated over $\tau' \approx 90$ days against the S&P 500 trend indicator, $\tau = 180$ days (the paper's Fig. 9). Dashed: quadratic fits.

Series	View	Sample	R^2 (quadratic fit)
Managed-futures fund	Monthly return vs S&P 500 monthly return	2010–2024	0.04
Managed-futures fund	P&L over $\tau' \approx 90$ d vs S&P 500 trend ($\tau = 180$ d)	2010–2024	0.10
Replicator (16 futures)	Monthly return vs S&P 500 monthly return	2003–2024	0.06
Replicator (16 futures)	P&L over $\tau' \approx 90$ d vs S&P 500 trend ($\tau = 180$ d)	2003–2024	0.09

Table 14: Convexity of the fund and of the 16-futures replicator against the S&P 500: naive monthly view vs aggregated view.

Measured the paper's way, the fund's convexity is clearer than in the naive view (R^2 0.04 $\rightarrow$ 0.10), and the replicator behaves the same way over 2003–2024 (0.06 $\rightarrow$ 0.09). The gain is about half the paper's (0.02 $\rightarrow$ 0.18 on the SG CTA Index, 2000–2015). The paper explains part of the gap: a diversified trend is convex in its own markets' moves, and only partly in the moves of one reference market such as the S&P 500.

10.4 A bound for risk-parity portfolios

The paper's answer to that dilution is eq. (24). Take the equal-risk long-only portfolio of the same markets, a simple risk-parity portfolio with daily return $G^{\text{RP}} = \sum_k w_k R_k$ and trend indicator $T_{\text{RP}} = \sqrt{\tau} L_\tau[G^{\text{RP}}]$. Because the square of an average is at most the average of the squares, the diversified trend's aggregated P&L can never fall below $\Upsilon(T_{\text{RP}}^2 - \sum_k w_k L_{\tau'}[R_k^2])$. This is a strict inequality, not a statistical tendency.

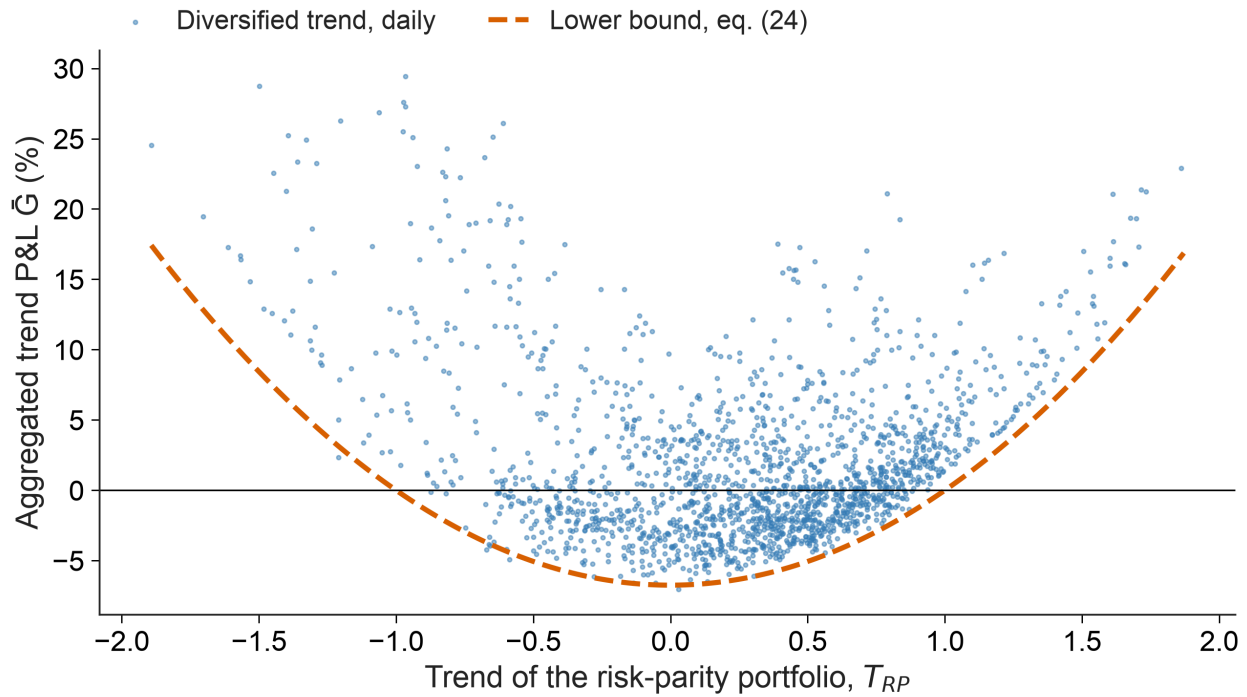


Figure 18: Aggregated P&L of the 16-futures trend against the trend of the equal-risk portfolio of the same futures, daily from 2003-05 (the paper's Fig. 10). Dashed: the bound with $\langle R^2 \rangle = 1$.

The exact bound holds on all 5434 days, as it must, and the simple parabola with $\langle R^2 \rangle = 1$ on 99.7% of them. Large moves of a risk-parity portfolio, up or down, over about six months are therefore always accompanied by trend profits. The inverse-volatility book below is a risk-parity portfolio of this kind, on three assets.

10.5 Trend as an overlay on the inverse-volatility book

The companion study builds a long-only book of the S&P 500, the 10-year Treasury and gold. It uses inverse-volatility weights from the trailing year, quarter-end rebalancing, 10 bp costs and a 10% ex-ante volatility target. The same rules are run here on futures from 1990, and a trend overlay is added:

- the paper's linear trend at $\tau = 180$ days on the 62 futures, or on the book's three assets only, or at $\tau = 40$ days;
- equal risk per market, next-close execution and the costs of Part II;
- the overlay is scaled to 10% ex-ante volatility, and every combination is rescaled to 10% ex-ante volatility, so that all portfolios carry the same risk.

Portfolio	Return p.a.	Vol	Sharpe	Max DD	Worst quarter	CVaR 5% (quarter)	Skew (quarter)
Book alone	7.2%	10.8%	0.66	-28%	-15.0%	-10.3%	-0.71
Book + 0.5 × diversified trend $\tau=180$	9.6%	10.3%	0.93	-21%	-16.9%	-10.0%	-0.24
Book + 1 × diversified trend $\tau=180$	10.2%	10.3%	0.99	-21%	-15.6%	-9.3%	0.12
Book + 1 × diversified trend $\tau=40$	6.6%	10.2%	0.64	-29%	-13.9%	-8.1%	0.45
Book + 1 × 3-asset trend $\tau=180$	7.0%	10.5%	0.67	-26%	-10.7%	-7.2%	0.60
Overlay alone, diversified trend $\tau=180$	7.8%	10.4%	0.75	-26%	-10.7%	-7.2%	0.84
Overlay alone, diversified trend $\tau=40$	1.9%	10.4%	0.18	-37%	-8.0%	-7.4%	1.19
Overlay alone, 3-asset trend $\tau=180$	4.0%	10.7%	0.37	-30%	-10.7%	-8.1%	0.58

Table 15: The inverse-vol book with and without a trend overlay, 1991-01 to 2024-03 combinations rescaled to 10% ex-ante volatility. Quarterly CVaR: mean of the worst 5% of quarters.

Episode	Book	Diversified, $\tau = 180$	Diversified, $\tau = 40$	Book's assets, $\tau = 180$
2000–02 dot-com bear	0.5%	52.9%	17.0%	29.5%
2007–09 financial crisis	8.8%	45.9%	46.5%	21.9%
2018 Q4 sell-off	-2.0%	-2.5%	3.9%	-3.7%
2020 COVID crash	-12.9%	8.1%	25.2%	3.0%
2022 rate shock	-26.7%	31.8%	13.3%	13.2%

Table 16: Returns over market stress episodes: the book, and each overlay on its own at 10% volatility.

At equal risk, the diversified slow trend improves the book on every summary measure except the single worst quarter (Table 15): Sharpe ratio $0.66 \rightarrow 0.99$, maximum drawdown $28\% \rightarrow 21\%$, quarterly skew $-0.71 \rightarrow 0.12$. Most of the gain comes from adding a positive-return stream with a low correlation to the book (0.19 monthly). Trend on the book's own three assets adds much less (Sharpe 0.67): diversification across markets matters more than hedging the same assets.

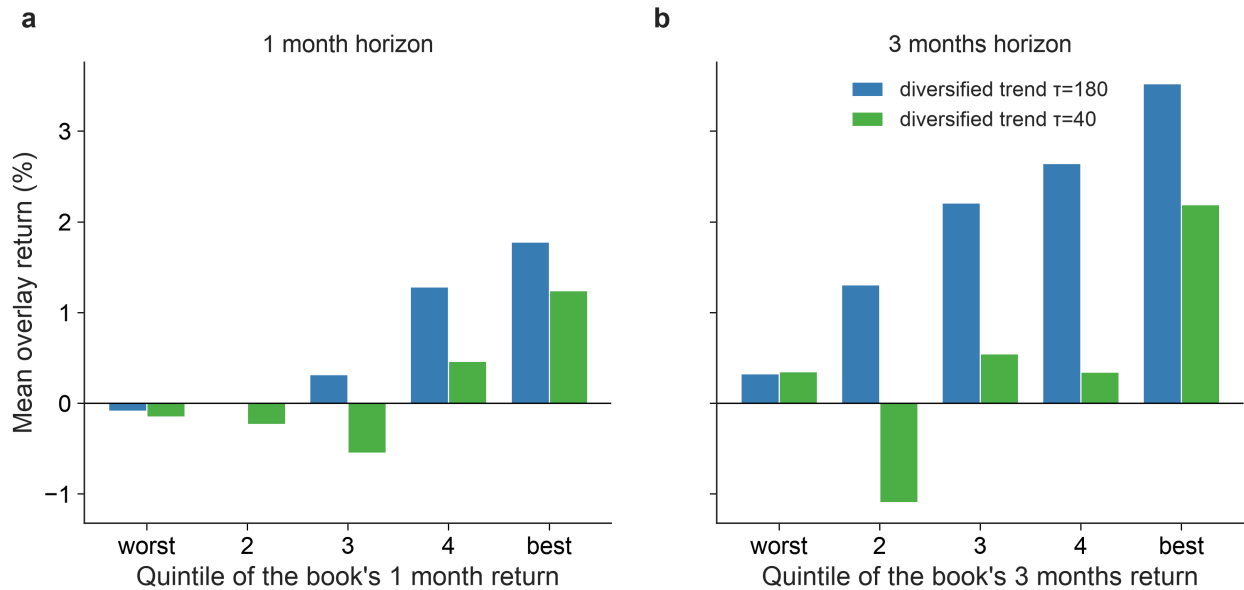


Figure 19: Mean return of the trend overlay by quintile of the book's return over the same month (a) or quarter (b).

The overlay is not a put on the book. On average it earns **more** when the book does well (Figure 19), because a trend follower is usually long the assets that are rising, and the book holds rising assets most of the time. Its protection is episodic and depends on time scale (Table 16). The slow trend made money through the slow bear markets: the 2000–02 dot-com bear, 2007–09 and the 2022 rate shock, when stocks and bonds fell together and the book lost 27%. In the five-week COVID crash the fast trend ($\tau = 40$) helped far more than the slow one, and the 2018 sell-off was too quick for the slow trend. This is the paper's warning made concrete: a six-month trend cannot hedge a crash that lasts a few weeks.

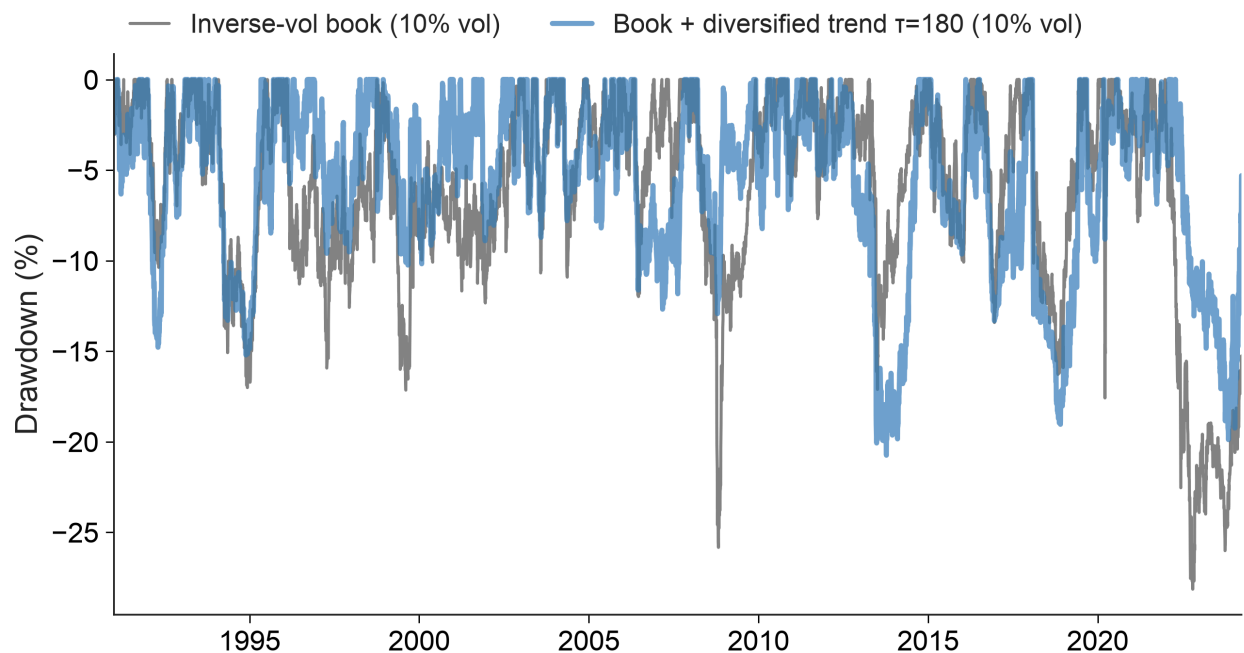


Figure 20: Drawdowns of the book alone and of the book with the diversified trend overlay, both at 10% volatility.

10.6 Trend vs options

The paper's last result is that a portfolio of strangles buys the same long-term variance as the trend, at a different price: the trend pays the short-term variance realised as it trades, the options pay the implied variance fixed at purchase. Options therefore protect against sudden moves that a trend cannot see, but they are sold at a premium. On the S&P 500, 1990–2024, the VIX (30-day implied volatility) averaged 21.1% against 18.3% realised over the following month, and exceeded it in 85% of months: implied variance cost 1.34 times realised variance.

Excess return over T-bills	Return p.a.	Vol	Sharpe	Max DD	2007–09	COVID 2020	2022
S&P 500	8.9%	18.1%	0.49	−60%	−56%	−34%	−25%
S&P 500 + 5% puts (CBOE PPUT)	5.3%	13.5%	0.39	−52%	−42%	−11%	−22%
S&P 500 + trend on the S&P 500	9.7%	21.5%	0.45	−60%	−45%	−38%	−26%
S&P 500 + diversified trend	17.0%	19.7%	0.86	−46%	−38%	−28%	−0%

Table 17: S&P 500 protection, 1990–2024: buying 5% out-of-the-money puts every month (CBOE PPUT index) vs adding a trend overlay at 10% volatility. Excess returns over T-bills; episodes as in Table 16.

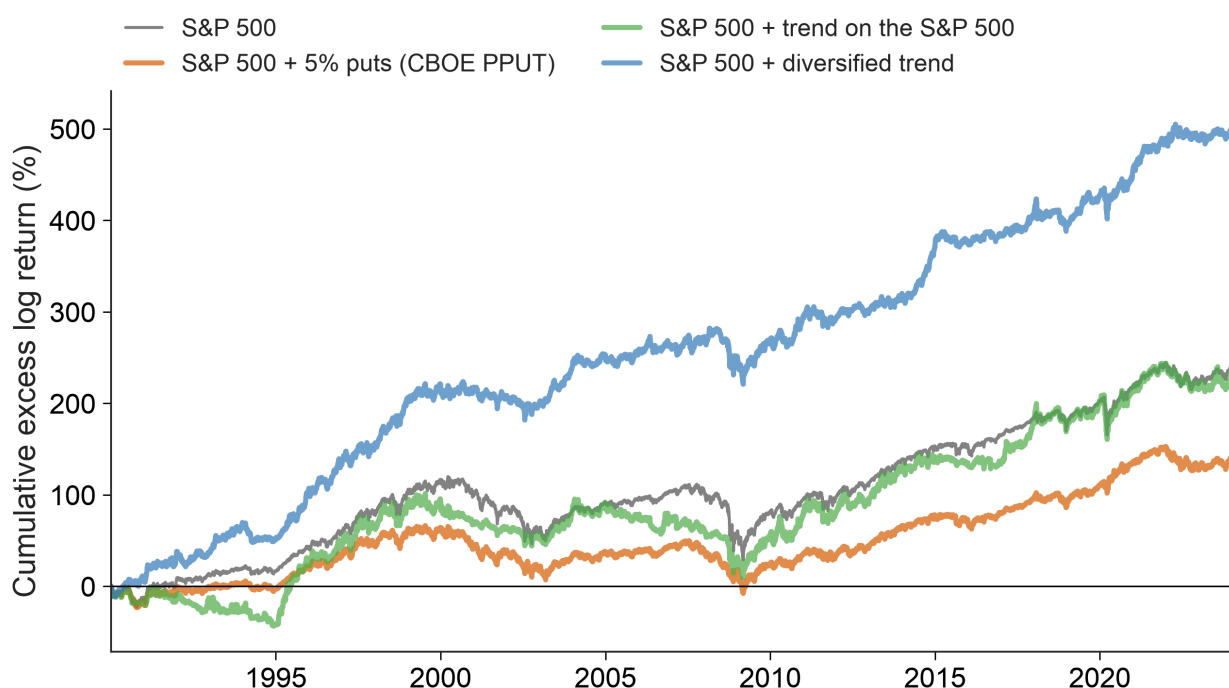


Figure 21: Cumulative excess log return of the S&P 500 alone, with monthly 5% puts, and with a trend overlay.

The put strategy cost 3.7% a year and lowered the Sharpe ratio, but it protected best in the COVID crash. Trend on the S&P 500 alone did little. The diversified trend overlay roughly doubled the excess return at slightly higher volatility, and protected in the slow 2007–09 and 2022 bear markets, much less in March 2020. These are not risk-matched portfolios, so the table shows trade-offs rather than a ranking. The pattern is the paper's: options are the better hedge, trend the cheaper one.

11 Conclusions and limitations

Part I. The paper's central claims replicate on free data: a significant, drift-independent trend effect across asset classes and decades, and saturation of the signal. Its size depends on details the paper does not dwell on. Monthly averages fake a trend unless the position is lagged, administered prices must be filtered, and interpolated history must be excluded. On spot data the effect weakened after publication to a Sharpe ratio of about 0.27, which is low but within what its own history allows.

Part II. On daily futures net of costs, Clenow's core model and the paper's signal earn similar Sharpe ratios (0.75 and 0.86 before 2014, 0.57 and 0.48 after) and are mostly the same trend exposure. Neither adds significant alpha to the other. The practitioner's stop does not add edge; it trades fewer days, with smaller drawdowns and more positive skew. The results hold across a wide range of parameters, but the best in-

sample settings did not stay best. This is consistent with the book's argument that diversification and risk sizing matter more than the exact rule, and with the paper's view of trend as a single robust effect: the book's rule and the paper's signal are two ways of trading it.

Part III. The convexity mechanism of Dao et al. replicates exactly where it is mathematical (the identity, the parabola and V on one market, the risk-parity bound), and more weakly where it is empirical: a public trend fund is convex in the S&P 500 at the right horizon, but less clearly than the SG CTA Index in the paper. For a long-only inverse-volatility book, a diversified slow trend overlay improves risk-adjusted returns, drawdowns and skew at equal risk. That protection is episodic and depends on time scale: it covers bear markets that unfold over months, such as 2022 when stocks and bonds fell together, and not crashes that last a few weeks. Puts cover those, at a cost of several percent a year.

Limitations. Part I: spot and index proxies, monthly averages, no costs. Part II: data end in March 2024; the universe is the set of contracts listed today, so it includes some survivorship and selection; ATR is estimated from closes; trades are at the next close, not the next open; costs are today's, scaled by volatility; positions are fractional and P&L is not compounded; and the managed-futures comparison uses a single fund. Part III: one public fund instead of the SG CTA Index; the convexity analysis uses gross P&L, as in the paper; the book is rebuilt on futures, not the ETF and UCITS implementation of the companion study; the put comparison uses one listed strategy and is not risk-matched.

Possible extensions. The SG CTA Index itself, if available; the convexity of Clenow's stop-based rules, whose positive skew (Part II) suggests a V-like profile; and a hedge built from actual option prices rather than an index.

12 Appendix: code and reproducibility

`uv run trendrep` (Part I), `uv run trendrep-daily` (Part II) and `uv run trendrep-convexity` (Part III, after Part II) download the raw data once (cached in `data/raw/`, not committed), run every analysis and write `results/` and `figures/`. `uv run trendrep-report` compiles this PDF, and `uv run pytest` runs the tests. The tests check the EMA and P&L against hand calculations and verify that there is no look-ahead (changing later prices leaves all earlier signals, positions and P&L unchanged). They also verify that random walks give t-stats centred on zero with unit spread, that monthly averaging fakes a trend at lag 0 but not at lag 1, that trending series are profitable, that stale prices are not traded, and that the saturation fit and the de-biasing recover known parameters. For Part II they check every entry, exit and holding day of the core model against its rules, that the filter blocks trades against the trend, the execution lag and P&L, and the cost accounting, and that the rules earn nothing on random walks and make money on persistent trends. For Part III they check the trend identity exactly on fat-tailed returns, the parabola and V-shape on random walks, the risk-parity bound on every day of correlated simulated returns, the volatility target of the book and that the risk estimates and the volatility scaling use past data only.

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